

Prometheus & Grafana Practical

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1. Monitor and create a dashboard of machine metrics like a CPU

- **Node Exporter** is a metrics collector for the machine. It is used to collect system-level metrics such as CPU, memory, disk, and network usage.
- Install Node Exporter on the Prometheus machine and start it in the background.

```
108.133]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 10676343 (10M) [application/octet-stream]
Saving to: 'node_exporter-1.8.2.linux-amd64.tar.gz'

node_exporter-1.8.2.linu 100%[=====>] 10.18M  --.-KB/s    in 0.04s

2026-02-13 09:30:38 (254 MB/s) - 'node_exporter-1.8.2.linux-amd64.tar.gz' saved [10676343/10676343]

root@ip-172-31-13-40:/home/ubuntu# ls
node_exporter-1.8.2.linux-amd64.tar.gz  prometheus-2.34.0.linux-amd64.tar.gz
prometheus-2.34.0.linux-amd64
root@ip-172-31-13-40:/home/ubuntu# sudo tar xvfz node_exporter-*.linux-amd64.tar.gz
node_exporter-1.8.2.linux-amd64/
node_exporter-1.8.2.linux-amd64/NOTICE
node_exporter-1.8.2.linux-amd64/node_exporter
node_exporter-1.8.2.linux-amd64/LICENSE
root@ip-172-31-13-40:/home/ubuntu# sudo rm node_exporter-1.8.2.linux-amd64.tar.gz
root@ip-172-31-13-40:/home/ubuntu# ls
node_exporter-1.8.2.linux-amd64  prometheus-2.34.0.linux-amd64.tar.gz
prometheus-2.34.0.linux-amd64
root@ip-172-31-13-40:/home/ubuntu# cd node_exporter-1.8.2.linux-amd64
root@ip-172-31-13-40:/home/ubuntu/node_exporter-1.8.2.linux-amd64# sudo ./node_exporter &
[1] 3568
root@ip-172-31-13-40:/home/ubuntu/node_exporter-1.8.2.linux-amd64#
```

- Configure Prometheus by updating the prometheus.yml file to scrape metrics from Node Exporter on port 9100.

```

# Load rules once and periodically evaluate them according to the global 'evaluation_interval'.
rule_files:
  # - "first_rules.yml"
  # - "second_rules.yml"

# A scrape configuration containing exactly one endpoint to scrape:
# Here it's Prometheus itself.
scrape_configs:
  # The job name is added as a label `job=<job_name>` to any timeseries scraped from this config.
  - job_name: "prometheus"

    # metrics_path defaults to '/metrics'
    # scheme defaults to 'http'.

    static_configs:
      - targets: ["localhost:9090"]

  - job_name: "docker"
    static_configs:
      - targets: ["localhost:9323"]

  - job_name: "node_exporter"
    static_configs:
      - targets: ["localhost:9100"]

-- INSERT --

```

35,2

Bot

- Restart Prometheus and verify that the Node Exporter target is **UP** in the Prometheus UI.

← → ↻ Not secure 51.20.115.124:9090/targets ☆ 🌧 📁 ⬇ 🔄 🔍 ⋮

Prometheus Alerts Graph Status ▾ Help ⚙ 🌙 🌕

Targets

All Unhealthy Collapse All 🔍 Filter by endpoint or labels

docker (1/1 up) [show less](#)

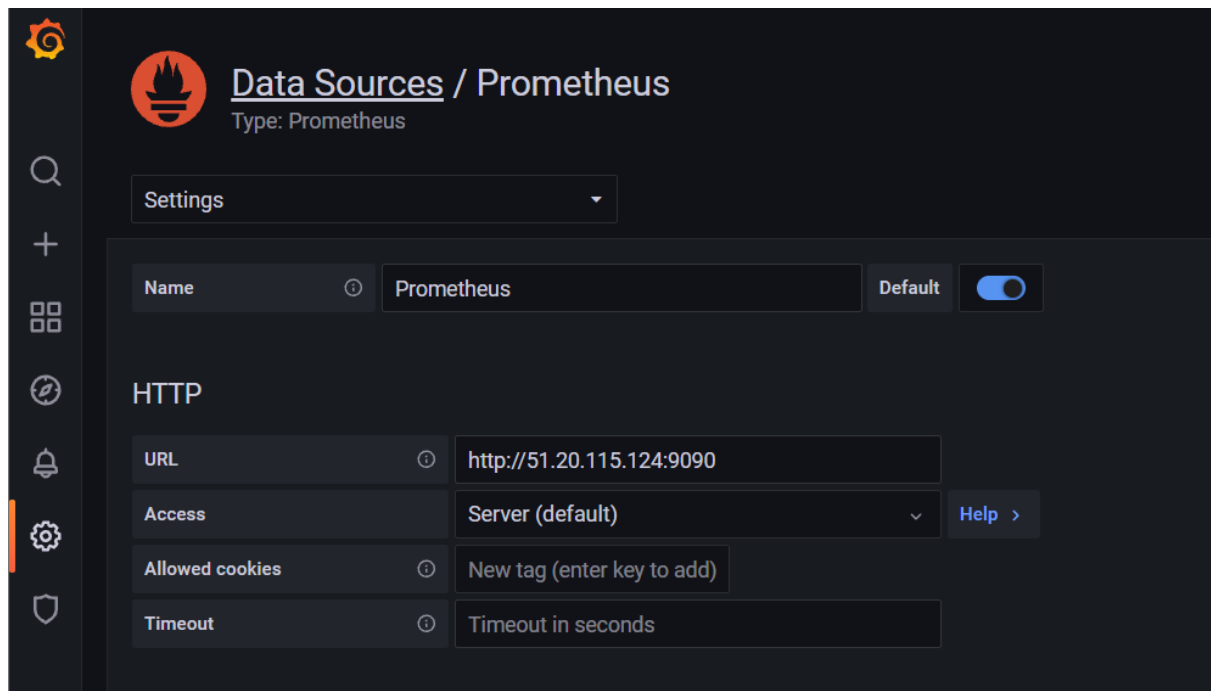
Endpoint	State	Labels	Last Scrape	Scrape Duration	Error
http://localhost:9323/metrics	UP	instance="localhost:9323" job="docker"	9.846s ago	7.483ms	

node_exporter (1/1 up) [show more](#)

prometheus (1/1 up) [show less](#)

Endpoint	State	Labels	Last Scrape	Scrape Duration	Error
http://localhost:9090/metrics	UP	instance="localhost:9090" job="prometheus"	486.000ms ago	6.831ms	

- Connect Grafana to Prometheus by adding Prometheus as a Data Source.



The screenshot shows the Grafana 'Data Sources / Prometheus' configuration page. The interface is dark-themed. On the left is a sidebar with navigation icons. The main content area has a header with the Prometheus logo and title. Below the title is a 'Settings' dropdown menu. The configuration is organized into sections: 'Name' (Prometheus, Default, and a toggle switch), 'HTTP' (URL: http://51.20.115.124:9090, Access: Server (default), Allowed cookies: New tag (enter key to add), Timeout: Timeout in seconds), and a 'Help' link.

Data Sources / Prometheus
Type: Prometheus

Settings

Name ⓘ Prometheus Default ☒

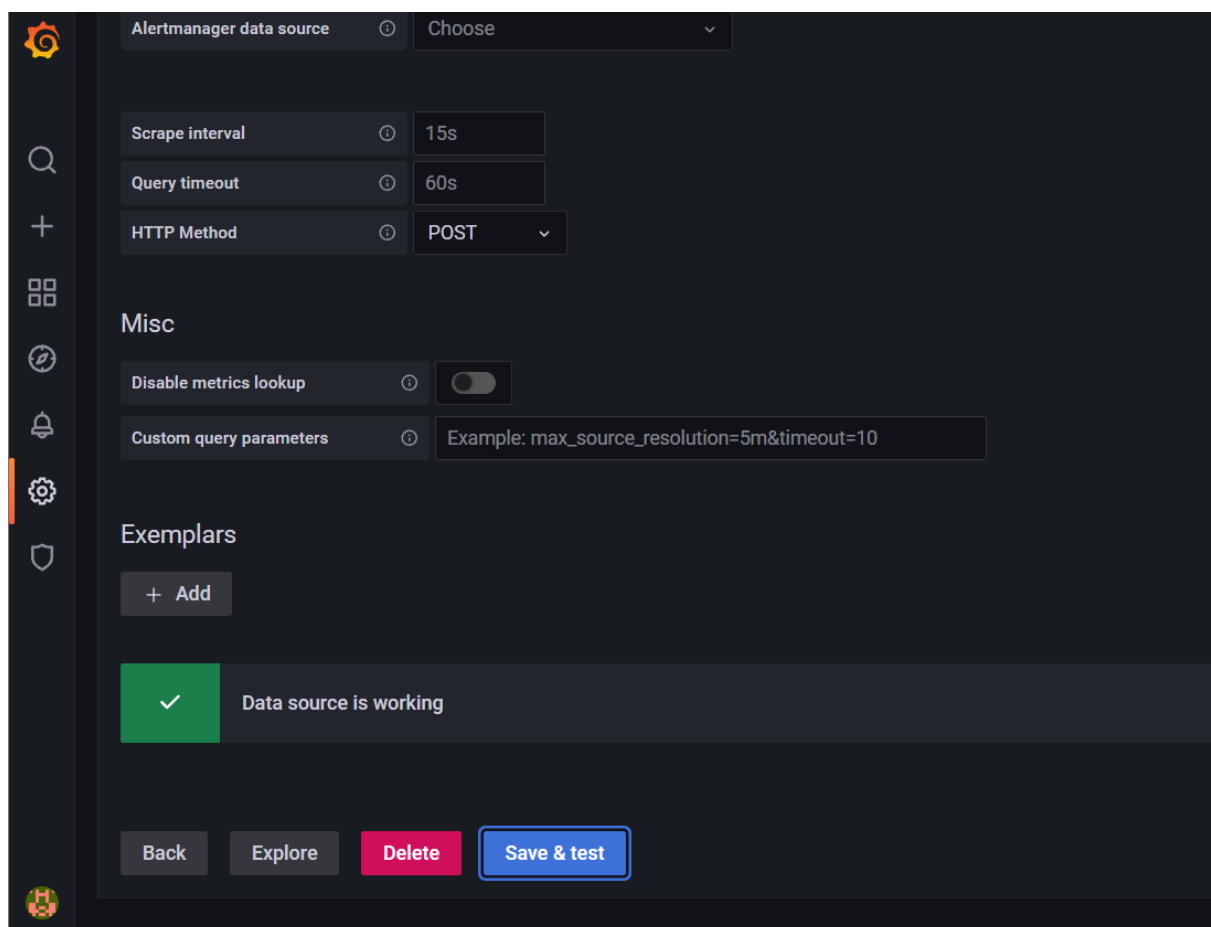
HTTP

URL ⓘ http://51.20.115.124:9090

Access Server (default) Help >

Allowed cookies ⓘ New tag (enter key to add)

Timeout ⓘ Timeout in seconds



The screenshot shows the Grafana 'Alertmanager data source' configuration page. The interface is dark-themed. On the left is a sidebar with navigation icons. The main content area has a header with the Alertmanager logo and title. Below the title is a 'Choose' dropdown menu. The configuration is organized into sections: 'Scrape interval' (15s), 'Query timeout' (60s), 'HTTP Method' (POST), 'Misc' (Disable metrics lookup: toggle switch, Custom query parameters: Example: max_source_resolution=5m&timeout=10), and 'Exemplars' (+ Add). At the bottom, there is a status bar with a green checkmark and the text 'Data source is working', and a row of buttons: Back, Explore, Delete, and Save & test.

Alertmanager data source ⓘ Choose

Scrape interval ⓘ 15s

Query timeout ⓘ 60s

HTTP Method ⓘ POST

Misc

Disable metrics lookup ⓘ ☐

Custom query parameters ⓘ Example: max_source_resolution=5m&timeout=10

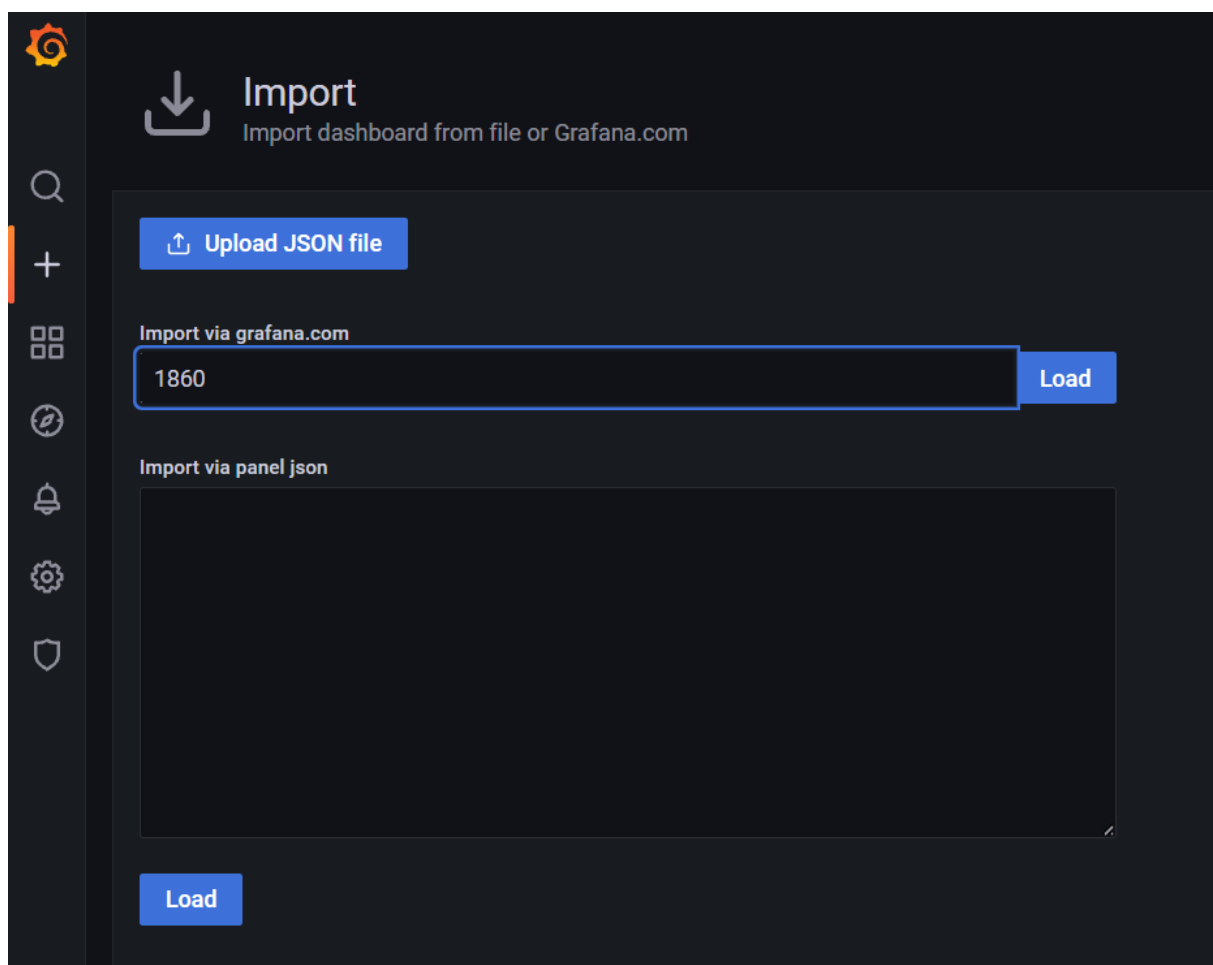
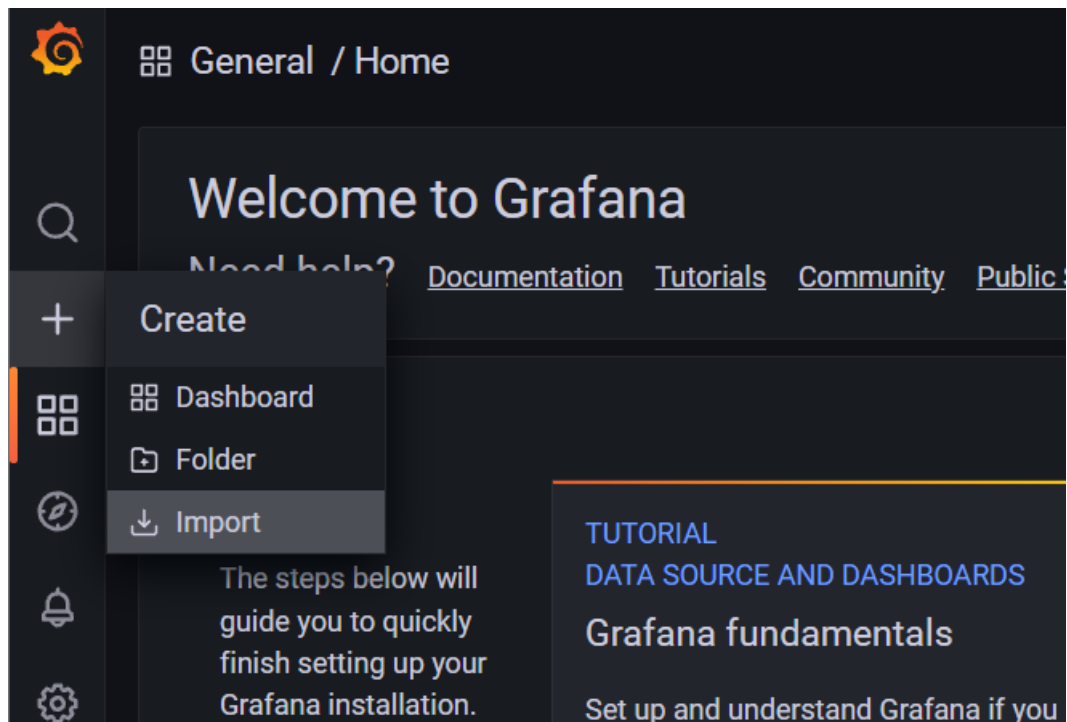
Exemplars

+ Add

✓ Data source is working

Back Explore Delete Save & test

- Import the Node Exporter Dashboard in Grafana using Dashboard ID 1860



- After importing the dashboard, system monitoring graphs such as CPU, memory, disk and network usage will be visible in Grafana.

